IMRAN ALI SOOMRO

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SUMMARY

Machine learning researcher and M.S. Software Engineering candidate focused on the robustness of image watermarking and content protection systems. This research designs and evaluates attacks that test how well current watermarking defenses hold up, an adversarial approach to stress testing the safeguards meant to protect images. First author on a submission to IEEE Transactions on Information Forensics and Security. Also built a research platform that coordinates several local and remote language models to run rigorous robustness evaluations at scale, spanning more than 100,000 images and 9 watermarking schemes. Technical foundation spans PyTorch, diffusion and generative models, and the orchestration of large language models. Seeking to apply this expertise to empirical AI safety research on the robustness, evaluation, and alignment of frontier AI systems.

RESEARCH INTERESTS

Image Watermarking and Attacks, Adversarial Robustness, Diffusion and Generative Models, Large Language Models and AI Agents, AI Safety and Model Evaluation, Computer Vision, Image Processing and Protection.

EDUCATION

M.S., Software Engineering — Hohai University Sep 2024 – Present
Nanjing, China · Advisor: Prof. Ping Ping · College of Computer Science and Software Engineering

B.E., Computer Systems Engineering — MUET Oct 2019 – Dec 2023
Jamshoro, Pakistan · Final grade: A+ · Thesis: HaptiCare — a VR-based therapy system (Best FYP Award)

PUBLICATIONS

Published

- T. A. Arain, P. C. Zhang, S. Mazhar, Q. Meng, **Imran Ali Soomro**, "Attention-guided multi-feature fusion hybrid architecture for person re-identification in smart surveillance systems," *Image and Vision Computing* (Elsevier), 2026. doi:10.1016/j.imavis.2026.106014
- M. M. Raza, **Imran Ali Soomro**, W. U. Khan, M. Q. Iqbal, "Generative Artificial Intelligence in Software Engineering: Redefining Programming Paradigms and Development Practices," *Spectrum of Engineering Sciences*, vol. 3, no. 10, pp. 335–348, 2025.

Under review

- **Imran Ali Soomro**, S. Abbas, S. Mazhar, A. Habib, T. A. Arain, A. Rahim, "SAG-SSNet: Scene-Adaptive Gated Spatiotemporal Segmentation Network for Robust Winter Road Segmentation," *Image and Vision Computing* (Elsevier), 2026.
- **Imran Ali Soomro**, A. Habib, S. Abbas, M. H. Zahoor, "A Residual Frequency–Temporal Network for Long-Horizon CPU Utilisation Forecasting in Cloud Data Centres," *Computing* (Springer).

Submitted

- **Imran Ali Soomro**, Ping Ping, "Erasing the Invisible: Anomaly-Gated Attention for Fidelity-Preserving Diffusion Watermark Removal," *IEEE Transactions on Information Forensics and Security*. *AI security / adversarial robustness*
- S. Abbas, S. Abbas, **Imran Ali Soomro**, M. Hussain, H. Zafar, Ahmad, "SPQG-YOLO26: Structure-Preserving Quality-Guided P3 Refinement for Lightweight Low-Light Object Detection," *Applied Soft Computing*.

RESEARCH EXPERIENCE

AI / ML Researcher — Hohai University Aug 2025 – Present
Nanjing, China

- Conduct novel research on diffusion-based image-watermarking attack mechanisms to evaluate the robustness of current defenses; first author on a submission to IEEE Transactions on Information Forensics and Security.
- Design reproducible experimental pipelines in PyTorch, run large-scale robustness evaluations with rigorous statistical analysis, and submit high-quality manuscripts to SCI journals (IEEE, Elsevier, Springer).

Research Assistant — Mehran University of Engineering & Technology Dec 2023 – May 2024

Jamshoro, Pakistan · Department of Computer Systems Engineering

- Supported the department through research collaboration, teaching, and administrative work as crash-semester coordinator.

SELECTED PROJECTS

InvisibleWatermarkLab — *multi-agent AI research platform for invisible-watermark robustness research*

- Orchestrated five LLMs (local: Qwen3.5-9B, Kimi-VL, gpt-oss-20b; remote: Nemotron, GLM) with automatic task-based routing; FastAPI backend, React 19 + TypeScript frontend.
- Built a Model Context Protocol (MCP) server integrating an external agent runtime (OpenClaw) with end-to-end cross-process tool discovery and a security-tested read-only data boundary.
- Engineered a process-safe GPU resource-lease system (cross-process file locking + PID-liveness detection) to prevent VRAM contention; verified with multi-process concurrency tests.
- Ran 8 large-scale experiments (100k+ generated / attacked images) evaluating 9 invisible-watermarking schemes against diffusion-reconstruction and geometric attacks, with bootstrap confidence intervals and paired Wilcoxon tests; 138+ automated tests.

Stack: Python, FastAPI, PyTorch, Diffusers / Transformers, React, TypeScript, Vite, MCP.

AutoLabler — *AI-powered full-stack image-annotation tool*

- FastAPI + React (Fabric.js canvas) unifying five vision models (YOLO11, YOLO-World, SAM, SAM2, DINOv3) behind one API with dynamic GPU load/unload; auto-labeling, interactive segmentation, and multi-format export (YOLO / COCO / VOC / LabelMe); Dockerized deployment.

FootballVision — *automated football-match analysis pipeline*

- Custom-trained YOLOv5 for player / ball / referee detection; per-player tracking with K-Means team assignment, optical-flow camera-motion compensation, and possession tracking; CLI + Streamlit app.

HaptiCare (Best FYP Award) — VR-based physiotherapy system with a haptic jacket for rotator-cuff rehabilitation.

Multimodal Brain-Stroke Prediction — ML multi-model system to detect & predict strokes in support of clinical decision-making.

Driver Drowsiness Detection — Real-time alertness monitor using a MobileNetV2 model deployed on a Raspberry Pi.

Smart Acoustical Eye — YOLOv8 object detection with spoken audio feedback to assist visually impaired users.

Snow-Road Slipperiness Detection — Real-time YOLOv8 segmentation of snow-covered roads with slippery-level classification.

Vehicle Tracking System — Real-time YOLOv8 vehicle detection, counting and multi-object tracking via Korean CCTV footage.

TECHNICAL SKILLS

ML / Deep Learning: PyTorch, TensorFlow, Keras; GANs & diffusion models, GenAI models, adversarial methods, transfer learning

LLMs / GenAI: LLMs, RAG, multi-agent orchestration, Model Context Protocol (MCP); Hugging Face Transformers & Diffusers; NLP, ViT, VLM, Dinov2/v3.

Computer Vision: OpenCV, YOLO (v5/11/World), SAM / SAM2, DINOv3, Faster R-CNN, Detectron, MediaPipe, OpenPose; detection, segmentation, tracking

Data / Scientific: NumPy, Pandas, SciPy, scikit-learn, Matplotlib; SQL; statistical testing (bootstrap CIs, Wilcoxon)

Engineering: Python, C, C++, C#, Java; FastAPI, React / TypeScript, Docker; Git, Linux; Raspberry Pi

HONOURS & AWARDS

Jiangsu Government Scholarship (2025); Hohai University Scholarship (2024); Best Final-Year Project Award, MUET (2023); 2nd Place & Silver Medal — Belt & Road Initiative O-Tour (2025).

CERTIFICATIONS & LANGUAGES

Certifications: Machine Learning Specialization — Stanford & DeepLearning.AI (Coursera); IBM Data Science Professional Certificate (Coursera); IELTS Academic; HSK 3 (Chinese)

Languages: Urdu (native); English (IELTS; CEFR C1–C2); Chinese (HSK 3)

Service: Volunteer, IEEE MUET Student Branch

REFERENCES

Prof. Ping Ping — Associate Professor, Hohai University (M.S. supervisor) · pingpingnjust@163.com

Dr. Shahnawaz Talpur — Associate Professor & Director of IT, MUET (B.E. supervisor) · shahnawaz.talpur@faculty.muet.edu.pk

Dr. Sanam Narejo — Associate Professor & Chairperson, MUET · sanam.narejo@faculty.muet.edu.pk